

| TARGET FILE      | TOTAL LINES | LANGUAGE   | DOMAIN                |
|------------------|-------------|------------|-----------------------|
| vector_search.py | 256 Lines   | Python 3.x | RAG / Codebase Search |

**Role in the project's RAG system:** This file is the **codebase retrieval** layer and works as a basic/fallback RAG search. `build_simple_rag_index()` walks a project folder, skips generated folders (`.git`, `node_modules`, `venv` and similar), reads supported files and stores each as an in-memory document (path, full content, tokens). `query_rag_engine()` then tokenizes the user's question, scores every file by token overlap, adds a boost for exact phrase matches, builds a snippet around the match and returns the top-k files. That retrieved code context is given to the generation/analysis stage so answers are grounded in the real project files. Despite the file name, it uses no embeddings or vector database, only word overlap, and needs no external libraries.

| #  | SOURCE CODE                                                        | TECHNICAL EXPLANATION                                                                                                                                                                |
|----|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <code>import os</code>                                             | Imports the operating-system interface module, used to walk folders, join/absolutize paths and check that directories exist.                                                         |
| 2  | <code>import re</code>                                             | Imports the regular-expression module, used by <code>_tokenize</code> to split text into searchable tokens.                                                                          |
| 3  |                                                                    | Blank spacer line separating the imports from the constants.                                                                                                                         |
| 4  |                                                                    | Second blank spacer line (PEP 8 spacing before a module-level section).                                                                                                              |
| 5  | <code># Files that are useful for codebase retrieval.</code>       | Comment: the list below holds the file types that are worth indexing when the RAG system searches a codebase.                                                                        |
| 6  | <code>SUPPORTED_EXTENSIONS = (</code>                              | Defines the constant tuple <code>SUPPORTED_EXTENSIONS</code> , the whitelist of file extensions that will be indexed. A tuple is used because <code>str.endswith</code> accepts one. |
| 7  | <code>" .py",</code>                                               | Allows Python source files ( <code>.py</code> ).                                                                                                                                     |
| 8  | <code>" .js",</code>                                               | Allows JavaScript files ( <code>.js</code> ).                                                                                                                                        |
| 9  | <code>" .jsx",</code>                                              | Allows React/JSX files ( <code>.jsx</code> ).                                                                                                                                        |
| 10 | <code>" .ts",</code>                                               | Allows TypeScript files ( <code>.ts</code> ).                                                                                                                                        |
| 11 | <code>" .tsx",</code>                                              | Allows TypeScript React files ( <code>.tsx</code> ).                                                                                                                                 |
| 12 | <code>" .md",</code>                                               | Allows Markdown documents ( <code>.md</code> ), such as README and docs, which carry useful project knowledge.                                                                       |
| 13 | <code>" .json",</code>                                             | Allows JSON data/config files ( <code>.json</code> ).                                                                                                                                |
| 14 | <code>" .txt",</code>                                              | Allows plain-text files ( <code>.txt</code> ).                                                                                                                                       |
| 15 | <code>)</code>                                                     | Closes the <code>SUPPORTED_EXTENSIONS</code> tuple.                                                                                                                                  |
| 16 |                                                                    | Blank spacer line between the constant and the first function.                                                                                                                       |
| 17 |                                                                    | Second blank spacer line (PEP 8: two blank lines before a top-level function).                                                                                                       |
| 18 | <code>def _tokenize(text):</code>                                  | Defines the private helper <code>_tokenize(text)</code> , the text-to-tokens step shared by indexing and searching.                                                                  |
| 19 | <code>    """Convert text into simple searchable tokens."""</code> | Docstring: converts text into simple searchable tokens.                                                                                                                              |
| 20 | <code>    if not text:</code>                                      | Guard clause: checks for empty or None input.                                                                                                                                        |

| #  | SOURCE CODE                                                            | TECHNICAL EXPLANATION                                                                                                                                                                                                      |
|----|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21 | <code>return []</code>                                                 | Returns an empty token list for empty input, so callers never receive None.                                                                                                                                                |
| 22 |                                                                        | Blank line separating the guard clause from the main logic.                                                                                                                                                                |
| 23 | <code>return re.findall(</code>                                        | Starts <code>re.findall(...)</code> , which returns every substring matching the pattern as a list.                                                                                                                        |
| 24 | <code>    r"[a-zA-Z0-9_]+",</code>                                     | The pattern <code>[a-zA-Z0-9_]+</code> matches runs of letters, digits and underscores. Keeping underscores means code identifiers such as <code>build_simple_rag_index</code> stay whole tokens, which suits code search. |
| 25 | <code>    str(text).lower()</code>                                     | Converts the input to a string and lowercases it, so matching is case-insensitive and non-string values do not crash the function.                                                                                         |
| 26 | <code>)</code>                                                         | Closes the <code>re.findall()</code> call; the resulting list is returned.                                                                                                                                                 |
| 27 |                                                                        | Blank spacer line between functions.                                                                                                                                                                                       |
| 28 |                                                                        | Second blank spacer line (PEP 8).                                                                                                                                                                                          |
| 29 | <code>def build_simple_rag_index(target_dir):</code>                   | Defines <code>build_simple_rag_index(target_dir)</code> , the indexing function of the RAG system. It reads the project's files once and returns them as a list of searchable documents.                                   |
| 30 | <code>    """</code>                                                   | Opens the multi-line docstring.                                                                                                                                                                                            |
| 31 | <code>    Read supported codebase files and create an in-memory</code> | Docstring: reads supported codebase files and creates an in-memory text index.                                                                                                                                             |
| 32 | <code>    text index.</code>                                           | Continues the docstring sentence: "text index".                                                                                                                                                                            |
| 33 |                                                                        | Blank line inside the docstring.                                                                                                                                                                                           |
| 34 | <code>    Each document stores:</code>                                 | Docstring: introduces what each stored document contains.                                                                                                                                                                  |
| 35 | <code>    - file path</code>                                           | Each document stores the file path.                                                                                                                                                                                        |
| 36 | <code>    - complete file content</code>                               | Each document stores the complete file content (used later to build snippets).                                                                                                                                             |
| 37 | <code>    - searchable tokens</code>                                   | Each document stores the searchable tokens (precomputed once, so searches are fast).                                                                                                                                       |
| 38 | <code>    """</code>                                                   | Closes the docstring.                                                                                                                                                                                                      |
| 39 |                                                                        | Blank line between the docstring and the code.                                                                                                                                                                             |
| 40 | <code>    documents = []</code>                                        | Creates the empty list <code>documents</code> that will collect one dictionary per indexed file.                                                                                                                           |
| 41 |                                                                        | Blank line before validation.                                                                                                                                                                                              |
| 42 | <code>    if not target_dir or not os.path.isdir(target_dir):</code>   | Guard: returns early if <code>target_dir</code> is empty/None or is not an existing directory.                                                                                                                             |
| 43 | <code>        return documents</code>                                  | Returns the empty <code>documents</code> list, so the caller receives a valid (empty) index instead of an error.                                                                                                           |
| 44 |                                                                        | Blank line before the folder walk.                                                                                                                                                                                         |
| 45 | <code>    for root, dirs, files in os.walk(target_dir):</code>         | Walks the folder tree recursively with <code>os.walk</code> . For every folder it yields the current path <code>root</code> , its sub-folders <code>dirs</code> and its file names <code>files</code> .                    |

| #  | SOURCE CODE                                                          | TECHNICAL EXPLANATION                                                                                                                                                                             |
|----|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 46 |                                                                      | Blank line at the start of the loop body.                                                                                                                                                         |
| 47 | <code># Avoid indexing generated/dependency directories.</code>      | Comment: avoid indexing generated or dependency directories, which would add noise and slow indexing.                                                                                             |
| 48 | <code>dirs[:] = [</code>                                             | Slice assignment <code>dirs[:] = [...]</code> edits the list in place. <code>os.walk</code> reads this list to decide which sub-folders to enter, so removed folders are never visited (pruning). |
| 49 | <code>    d for d in dirs</code>                                     | List comprehension: keeps each directory name <code>d</code> from <code>dirs</code> ...                                                                                                           |
| 50 | <code>    if d not in {</code>                                       | ... only if it is not in the exclusion set below. A set gives fast membership checks.                                                                                                             |
| 51 | <code>        ".git",</code>                                         | Excludes <code>.git</code> (version-control internals).                                                                                                                                           |
| 52 | <code>        "__pycache__",</code>                                  | Excludes <code>__pycache__</code> (compiled Python bytecode).                                                                                                                                     |
| 53 | <code>        ".venv",</code>                                        | Excludes <code>.venv</code> (virtual environment).                                                                                                                                                |
| 54 | <code>        "venv",</code>                                         | Excludes <code>venv</code> (virtual environment, alternate name).                                                                                                                                 |
| 55 | <code>        "node_modules",</code>                                 | Excludes <code>node_modules</code> (JavaScript dependencies).                                                                                                                                     |
| 56 | <code>        ".streamlit",</code>                                   | Excludes <code>.streamlit</code> (Streamlit config/cache folder).                                                                                                                                 |
| 57 | <code>    }</code>                                                   | Closes the exclusion set.                                                                                                                                                                         |
| 58 | <code>]</code>                                                       | Closes the list comprehension and the assignment.                                                                                                                                                 |
| 59 |                                                                      | Blank line before the file loop.                                                                                                                                                                  |
| 60 | <code>for file in files:</code>                                      | Loops over each file name in the current folder.                                                                                                                                                  |
| 61 |                                                                      | Blank line at the start of the inner loop body.                                                                                                                                                   |
| 62 | <code>    if not file.lower().endswith(SUPPORTED_EXTENSIONS):</code> | Checks whether the lower-cased file name ends with any supported extension ( <code>endswith</code> accepts a tuple). Unsupported files fail this test.                                            |
| 63 | <code>        continue</code>                                        | Skips unsupported files and moves to the next file.                                                                                                                                               |
| 64 |                                                                      | Blank line before path creation.                                                                                                                                                                  |
| 65 | <code>    file_path = os.path.abspath(</code>                        | Starts building the file's absolute path with <code>os.path.abspath(...)</code> .                                                                                                                 |
| 66 | <code>        os.path.join(root, file)</code>                        | Joins the folder <code>root</code> with the <code>file</code> name using the correct path separator for the OS.                                                                                   |
| 67 | <code>    )</code>                                                   | Closes the <code>abspath()</code> call, so <code>file_path</code> is a full, unambiguous path.                                                                                                    |
| 68 |                                                                      | Blank line before the file reading.                                                                                                                                                               |
| 69 | <code>    try:</code>                                                | Starts a <code>try</code> block so one unreadable file cannot crash the whole indexing run.                                                                                                       |
| 70 | <code>        with open(</code>                                      | Opens a <code>with open()</code> block (the file is closed automatically). Arguments follow on separate lines.                                                                                    |
| 71 | <code>            file_path,</code>                                  | First argument: the file path to open.                                                                                                                                                            |

| #  | SOURCE CODE                                                             | TECHNICAL EXPLANATION                                                                                                                         |
|----|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 72 | <code>"r",</code>                                                       | Mode "r": read as text.                                                                                                                       |
| 73 | <code>encoding="utf-8",</code>                                          | Encoding UTF-8, the standard for source code.                                                                                                 |
| 74 | <code>errors="ignore"</code>                                            | <code>errors="ignore"</code> drops undecodable bytes instead of raising an exception, which makes the reader tolerant of odd files.           |
| 75 | <code>) as f:</code>                                                    | Closes the <code>open()</code> call and binds the file object to <code>f</code> .                                                             |
| 76 | <code>content = f.read()</code>                                         | Reads the entire file into the string <code>content</code> .                                                                                  |
| 77 |                                                                         | Blank line after reading.                                                                                                                     |
| 78 | <code>if not content.strip():</code>                                    | Checks whether the file is empty or contains only whitespace ( <code>strip()</code> removes it).                                              |
| 79 | <code>continue</code>                                                   | Skips empty files, since they add nothing to the index.                                                                                       |
| 80 |                                                                         | Blank line before adding the document.                                                                                                        |
| 81 | <code>documents.append({</code>                                         | Appends a new document (a dictionary) to the index list <code>documents</code> .                                                              |
| 82 | <code>"path": file_path,</code>                                         | Key <code>path</code> : the absolute path of the file, returned later as the search result's source.                                          |
| 83 | <code>"content": content,</code>                                        | Key <code>content</code> : the full text of the file, used for snippets and phrase matching.                                                  |
| 84 | <code>"tokens": _tokenize(content),</code>                              | Key <code>tokens</code> : the tokenized content, computed once at index time so searches do not re-tokenize.                                  |
| 85 | <code>})</code>                                                         | Closes the dictionary and the <code>append()</code> call.                                                                                     |
| 86 |                                                                         | Blank line before the exception handler.                                                                                                      |
| 87 | <code>except (OSError, UnicodeError):</code>                            | Catches file-system errors ( <code>OSError</code> ) and text-decoding errors ( <code>UnicodeError</code> ).                                   |
| 88 | <code># Skip unreadable files without crashing.</code>                  | Comment: skip unreadable files without crashing.                                                                                              |
| 89 | <code>continue</code>                                                   | Continues with the next file, leaving the bad one out of the index.                                                                           |
| 90 |                                                                         | Blank line after the loops.                                                                                                                   |
| 91 | <code>return documents</code>                                           | Returns the completed index (list of documents). This is the knowledge base that <code>query_rag_engine</code> searches.                      |
| 92 |                                                                         | Blank spacer line between functions.                                                                                                          |
| 93 |                                                                         | Second blank spacer line (PEP 8).                                                                                                             |
| 94 | <code>def _calculate_match_score(query_tokens, document_tokens):</code> | Defines <code>_calculate_match_score</code> , the scoring function of the retriever. It compares the query tokens with one document's tokens. |
| 95 | <code>"""</code>                                                        | Opens the docstring.                                                                                                                          |
| 96 | <code>Calculate a simple token-overlap score.</code>                    | Docstring: calculates a simple token-overlap score.                                                                                           |
| 97 |                                                                         | Blank line inside the docstring.                                                                                                              |

| #   | SOURCE CODE                                                         | TECHNICAL EXPLANATION                                                                                                                                                                                |
|-----|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 98  | <code>This is intentionally lightweight and dependency-free.</code> | Docstring: the method is intentionally lightweight and dependency-free.                                                                                                                              |
| 99  | <code>It is not a neural embedding model.</code>                    | Docstring: clarifies it is not a neural embedding model. Despite the file name <code>vector_search.py</code> , no embeddings or vectors are used, only word overlap.                                 |
| 100 | <code>"""</code>                                                    | Closes the docstring.                                                                                                                                                                                |
| 101 |                                                                     | Blank line before the guard.                                                                                                                                                                         |
| 102 | <code>if not query_tokens or not document_tokens:</code>            | Guard clause: checks whether either token list is empty.                                                                                                                                             |
| 103 | <code>return 0.0</code>                                             | Returns a score of 0.0 (no match) in that case.                                                                                                                                                      |
| 104 |                                                                     | Blank line before building the lookup set.                                                                                                                                                           |
| 105 | <code>document_token_set = set(document_tokens)</code>              | Converts the document tokens into a set, so each "is this word in the document?" test is fast (constant time).                                                                                       |
| 106 |                                                                     | Blank line before counting matches.                                                                                                                                                                  |
| 107 | <code>matched_tokens = sum(</code>                                  | Starts <code>matched_tokens = sum(...)</code> over a generator expression.                                                                                                                           |
| 108 | <code>1</code>                                                      | Adds 1 for each matching token.                                                                                                                                                                      |
| 109 | <code>for token in set(query_tokens)</code>                         | Loops over the unique query tokens ( <code>set(query_tokens)</code> ), so a word repeated in the query is counted only once.                                                                         |
| 110 | <code>if token in document_token_set</code>                         | Counts the token only if it appears in the document's token set.                                                                                                                                     |
| 111 | <code>)</code>                                                      | Closes the <code>sum()</code> call. <code>matched_tokens</code> is how many distinct query words the document contains.                                                                              |
| 112 |                                                                     | Blank line before the denominator.                                                                                                                                                                   |
| 113 | <code>unique_query_tokens = len(set(query_tokens))</code>           | Counts the number of unique tokens in the query (the maximum possible matches).                                                                                                                      |
| 114 |                                                                     | Blank line before the safety check.                                                                                                                                                                  |
| 115 | <code>if unique_query_tokens == 0:</code>                           | Extra safety check against division by zero.                                                                                                                                                         |
| 116 | <code>return 0.0</code>                                             | Returns 0.0 if the query has no unique tokens.                                                                                                                                                       |
| 117 |                                                                     | Blank line before the return.                                                                                                                                                                        |
| 118 | <code>return matched_tokens / unique_query_tokens</code>            | Returns <code>matched_tokens / unique_query_tokens</code> , the fraction of query words found in the document (0.0 to 1.0). 1.0 means every query word appears in the file.                          |
| 119 |                                                                     | Blank spacer line between functions.                                                                                                                                                                 |
| 120 |                                                                     | Second blank spacer line (PEP 8).                                                                                                                                                                    |
| 121 | <code>def _make_snippet(content, query, window=180):</code>         | Defines <code>_make_snippet(content, query, window=180)</code> , which builds a short preview of the matching file. <code>window</code> is how many characters to include on each side of the match. |
| 122 | <code>"""</code>                                                    | Opens the docstring.                                                                                                                                                                                 |
| 123 | <code>Create a useful snippet around the first matching</code>      | Docstring: creates a useful snippet around the first matching query phrase or token.                                                                                                                 |

| #   | SOURCE CODE                                                 | TECHNICAL EXPLANATION                                                                                            |
|-----|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 124 | <code>query phrase or token.</code>                         | Continues the docstring sentence.                                                                                |
| 125 | <code>"""</code>                                            | Closes the docstring.                                                                                            |
| 126 |                                                             | Blank line before the guard.                                                                                     |
| 127 | <code>if not content:</code>                                | Guard: checks for empty content.                                                                                 |
| 128 | <code>return ""</code>                                      | Returns an empty string when there is no content.                                                                |
| 129 |                                                             | Blank line before preparing lowercase copies.                                                                    |
| 130 | <code>content_lower = content.lower()</code>                | Creates a lowercase copy of the file content for case-insensitive searching.                                     |
| 131 | <code>query_lower = query.lower()</code>                    | Creates a lowercase copy of the query.                                                                           |
| 132 |                                                             | Blank line before the phrase search.                                                                             |
| 133 | <code># First try the complete query.</code>                | Comment: strategy 1, try the complete query as an exact phrase first.                                            |
| 134 | <code>exact_index = content_lower.find(query_lower)</code>  | Uses <code>str.find</code> to get the position of the full query phrase in the content; returns -1 if not found. |
| 135 |                                                             | Blank line before the branch.                                                                                    |
| 136 | <code>if exact_index &gt;= 0:</code>                        | Checks whether the exact phrase was found (index 0 or greater).                                                  |
| 137 | <code>start = max(0, exact_index - window)</code>           | Computes the snippet start: the match position minus window, but never below 0 (max).                            |
| 138 | <code>end = min(</code>                                     | Starts computing the snippet end with <code>min(...)</code> .                                                    |
| 139 | <code>len(content),</code>                                  | First candidate: the file length, so the slice cannot run past the end of the file.                              |
| 140 | <code>exact_index + len(query) + window</code>              | Second candidate: match position + query length + window, i.e. the window after the phrase.                      |
| 141 | <code>)</code>                                              | Closes the <code>min()</code> call; end is the smaller of the two values.                                        |
| 142 |                                                             | Blank line before the return.                                                                                    |
| 143 | <code>return (</code>                                       | Starts the returned expression (parenthesised so it can be split across lines).                                  |
| 144 | <code>content[start:end]</code>                             | Slices the original content from <code>start</code> to <code>end</code> to get the snippet.                      |
| 145 | <code>.replace("\n", " ")</code>                            | Replaces newline characters with spaces so the snippet is a single readable line.                                |
| 146 | <code>.strip()</code>                                       | Trims leading and trailing whitespace.                                                                           |
| 147 | <code>)</code>                                              | Closes the parentheses; the cleaned snippet is returned.                                                         |
| 148 |                                                             | Blank line between strategy 1 and strategy 2.                                                                    |
| 149 | <code># Otherwise find the first useful query token.</code> | Comment: strategy 2, if the whole phrase is not found, look for the first individual query token.                |
| 150 | <code>for token in _tokenize(query):</code>                 | Loops over the tokens of the query, in order.                                                                    |
| 151 |                                                             | Blank line at the start of the loop body.                                                                        |

| #   | SOURCE CODE                                                      | TECHNICAL EXPLANATION                                                                                                                                                                       |
|-----|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 152 | <code>index = content_lower.find(token.lower())</code>           | Finds the position of the current token in the lowercase content.                                                                                                                           |
| 153 |                                                                  | Blank line before the check.                                                                                                                                                                |
| 154 | <code>if index &gt;= 0:</code>                                   | Checks whether the token was found.                                                                                                                                                         |
| 155 | <code>start = max(0, index - window)</code>                      | Computes the snippet start around the token position, clamped to 0.                                                                                                                         |
| 156 | <code>end = min(</code>                                          | Starts computing the snippet end with <code>min(...)</code> .                                                                                                                               |
| 157 | <code>len(content),</code>                                       | Upper bound: the file length.                                                                                                                                                               |
| 158 | <code>index + len(token) + window</code>                         | Token position + token length + window.                                                                                                                                                     |
| 159 | <code>)</code>                                                   | Closes the <code>min()</code> call.                                                                                                                                                         |
| 160 |                                                                  | Blank line before the return.                                                                                                                                                               |
| 161 | <code>return (</code>                                            | Starts the returned expression.                                                                                                                                                             |
| 162 | <code>content[start:end]</code>                                  | Slices the content around the matched token.                                                                                                                                                |
| 163 | <code>.replace("\n", " ")</code>                                 | Replaces newlines with spaces.                                                                                                                                                              |
| 164 | <code>.strip()</code>                                            | Trims whitespace.                                                                                                                                                                           |
| 165 | <code>)</code>                                                   | Closes the parentheses; returns the token-based snippet at the first token that matches.                                                                                                    |
| 166 |                                                                  | Blank line between strategy 2 and the fallback.                                                                                                                                             |
| 167 | <code># No direct token found.</code>                            | Comment: fallback, no direct token was found.                                                                                                                                               |
| 168 | <code>return content[:window * 2].replace(</code>                | Returns the beginning of the file: <code>content[:window * 2]</code> (360 characters by default) with newlines replaced.                                                                    |
| 169 | <code>"\n",</code>                                               | First argument of <code>replace</code> : the newline character to remove.                                                                                                                   |
| 170 | <code>" "</code>                                                 | Second argument: a single space to put in its place.                                                                                                                                        |
| 171 | <code>).strip()</code>                                           | Closes <code>replace()</code> and applies <code>.strip()</code> to the result before returning it.                                                                                          |
| 172 |                                                                  | Blank spacer line between functions.                                                                                                                                                        |
| 173 |                                                                  | Second blank spacer line (PEP 8).                                                                                                                                                           |
| 174 | <code>def query_rag_engine(documents, query, top_k=5):</code>    | Defines <code>query_rag_engine(documents, query, top_k=5)</code> , the public search function. It takes the index built earlier, a query string and how many results to return (default 5). |
| 175 | <code>"""</code>                                                 | Opens the docstring.                                                                                                                                                                        |
| 176 | <code>Search indexed codebase documents using lightweight</code> | Docstring: searches indexed codebase documents using lightweight token matching.                                                                                                            |
| 177 | <code>token matching and return the best matching files.</code>  | Docstring: returns the best matching files.                                                                                                                                                 |
| 178 |                                                                  | Blank line inside the docstring.                                                                                                                                                            |

| #   | SOURCE CODE                                                       | TECHNICAL EXPLANATION                                                                                                          |
|-----|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 179 | <code>This is a dependency-free retrieval layer and can be</code> | Docstring: this is a dependency-free retrieval layer.                                                                          |
| 180 | <code>used as a basic/fallback RAG search.</code>                 | Docstring: it can be used as a basic or fallback RAG search, for example when a heavier vector-based retriever is unavailable. |
| 181 | <code>"""</code>                                                  | Closes the docstring.                                                                                                          |
| 182 |                                                                   | Blank line before validation.                                                                                                  |
| 183 | <code>if not documents:</code>                                    | Checks whether the index is empty or None.                                                                                     |
| 184 | <code>return []</code>                                            | Returns an empty list, since there is nothing to search.                                                                       |
| 185 |                                                                   | Blank line between checks.                                                                                                     |
| 186 | <code>if not query or not str(query).strip():</code>              | Checks whether the query is empty, None or only whitespace.                                                                    |
| 187 | <code>return []</code>                                            | Returns an empty list for an empty query.                                                                                      |
| 188 |                                                                   | Blank line before query normalization.                                                                                         |
| 189 | <code>query = str(query).strip()</code>                           | Normalizes the query: converts it to a string and strips surrounding whitespace.                                               |
| 190 | <code>query_tokens = _tokenize(query)</code>                      | Tokenizes the query with the same <code>_tokenize</code> used at index time, so both sides use identical rules.                |
| 191 |                                                                   | Blank line before the check.                                                                                                   |
| 192 | <code>if not query_tokens:</code>                                 | Checks whether the query produced no tokens (for example, only symbols).                                                       |
| 193 | <code>return []</code>                                            | Returns an empty list in that case.                                                                                            |
| 194 |                                                                   | Blank line before <code>top_k</code> handling.                                                                                 |
| 195 | <code># Safely normalize top_k.</code>                            | Comment: safely normalize <code>top_k</code> .                                                                                 |
| 196 | <code>try:</code>                                                 | Starts a try block for the integer conversion.                                                                                 |
| 197 | <code>top_k = int(top_k)</code>                                   | Converts <code>top_k</code> to an integer (handles values such as "3" or 3.0).                                                 |
| 198 | <code>except (TypeError, ValueError):</code>                      | Catches <code>TypeError</code> (None, wrong type) and <code>ValueError</code> (non-numeric text).                              |
| 199 | <code>top_k = 5</code>                                            | Falls back to the default of 5 results.                                                                                        |
| 200 |                                                                   | Blank line before the range check.                                                                                             |
| 201 | <code>if top_k &lt;= 0:</code>                                    | Checks whether <code>top_k</code> is zero or negative.                                                                         |
| 202 | <code>return []</code>                                            | Returns an empty list, because zero results were requested.                                                                    |
| 203 |                                                                   | Blank line before the result containers.                                                                                       |
| 204 | <code>scored_results = []</code>                                  | Creates the empty list <code>scored_results</code> that will hold every matching file with its score.                          |
| 205 | <code>seen_paths = set()</code>                                   | Creates the empty set <code>seen_paths</code> , used to detect duplicate files.                                                |

| #   | SOURCE CODE                                                  | TECHNICAL EXPLANATION                                                                                                    |
|-----|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| 206 |                                                              | Blank line before the main loop.                                                                                         |
| 207 | <code>for doc in documents:</code>                           | Loops over every document in the index.                                                                                  |
| 208 |                                                              | Blank line at the start of the loop body.                                                                                |
| 209 | <code>if not isinstance(doc, dict):</code>                   | Checks that the item is a dictionary, which protects against malformed index entries.                                    |
| 210 | <code>continue</code>                                        | Skips anything that is not a dictionary.                                                                                 |
| 211 |                                                              | Blank line before reading fields.                                                                                        |
| 212 | <code>content = doc.get("content", "")</code>                | Reads the file content from the document (default is an empty string).                                                   |
| 213 | <code>path = doc.get("path", "")</code>                      | Reads the file path from the document (default is an empty string).                                                      |
| 214 |                                                              | Blank line before the check.                                                                                             |
| 215 | <code>if not content or not path:</code>                     | Checks whether the content or the path is missing or empty.                                                              |
| 216 | <code>continue</code>                                        | Skips such incomplete documents.                                                                                         |
| 217 |                                                              | Blank line before de-duplication.                                                                                        |
| 218 | <code># Prevent duplicate files from appearing twice.</code> | Comment: prevent the same file from appearing twice in the results.                                                      |
| 219 | <code>normalized_path = os.path.abspath(path)</code>         | Converts the path to an absolute, normalized form so the same file written in different ways is recognized as identical. |
| 220 |                                                              | Blank line before the duplicate check.                                                                                   |
| 221 | <code>if normalized_path in seen_paths:</code>               | Checks whether this path was already processed.                                                                          |
| 222 | <code>continue</code>                                        | Skips the duplicate document.                                                                                            |
| 223 |                                                              | Blank line before recording the path.                                                                                    |
| 224 | <code>seen_paths.add(normalized_path)</code>                 | Adds the path to <code>seen_paths</code> so later duplicates are skipped.                                                |
| 225 |                                                              | Blank line before token lookup.                                                                                          |
| 226 | <code>document_tokens = doc.get("tokens")</code>             | Reads the precomputed tokens stored at index time (None if absent).                                                      |
| 227 |                                                              | Blank line before the check.                                                                                             |
| 228 | <code>if document_tokens is None:</code>                     | Checks whether the document has no stored tokens.                                                                        |
| 229 | <code>document_tokens = _tokenize(content)</code>            | Tokenizes the content on the fly as a fallback, so documents built without tokens still work.                            |
| 230 |                                                              | Blank line before scoring.                                                                                               |
| 231 | <code>score = _calculate_match_score(</code>                 | Calls <code>_calculate_match_score(...)</code> to score this document against the query.                                 |
| 232 | <code>query_tokens,</code>                                   | Argument 1: the query tokens.                                                                                            |

| #   | SOURCE CODE                                                | TECHNICAL EXPLANATION                                                                                                                                                                              |
|-----|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 233 | <code>document_tokens</code>                               | Argument 2: the document tokens.                                                                                                                                                                   |
| 234 | <code>)</code>                                             | Closes the call; <code>score</code> is the fraction of query words found (0.0 to 1.0).                                                                                                             |
| 235 |                                                            | Blank line before the phrase boost.                                                                                                                                                                |
| 236 | <code># Give an exact phrase match a ranking boost.</code> | Comment: an exact phrase match gets a ranking boost.                                                                                                                                               |
| 237 | <code>if query.lower() in content.lower():</code>          | Checks whether the full query text appears as-is inside the file (case-insensitive).                                                                                                               |
| 238 | <code>score += 1.0</code>                                  | Adds 1.0 to the score. Files containing the exact phrase therefore always outrank files that only share scattered words (the maximum score becomes 2.0).                                           |
| 239 |                                                            | Blank line before the relevance check.                                                                                                                                                             |
| 240 | <code>if score &gt; 0:</code>                              | Keeps only relevant files: a score greater than 0 means at least one query word matched.                                                                                                           |
| 241 | <code>scored_results.append({</code>                       | Appends a result dictionary to <code>scored_results</code> (opens <code>append({</code> ).                                                                                                         |
| 242 | <code>"path": normalized_path,</code>                      | Result key <code>path</code> : the normalized absolute file path.                                                                                                                                  |
| 243 | <code>"snippet": _make_snippet(</code>                     | Result key <code>snippet</code> : calls <code>_make_snippet(</code> to generate a short preview of the matching area.                                                                              |
| 244 | <code>content,</code>                                      | Snippet argument 1: the full file content.                                                                                                                                                         |
| 245 | <code>query</code>                                         | Snippet argument 2: the query.                                                                                                                                                                     |
| 246 | <code>),</code>                                            | Closes the <code>_make_snippet(</code> call.                                                                                                                                                       |
| 247 | <code>"score": round(score, 4),</code>                     | Result key <code>score</code> : the score rounded to 4 decimal places for a clean output.                                                                                                          |
| 248 | <code>})</code>                                            | Closes the result dictionary and the <code>append(</code> call.                                                                                                                                    |
| 249 |                                                            | Blank line after the loop.                                                                                                                                                                         |
| 250 | <code># Highest-scoring documents first.</code>            | Comment: highest-scoring documents first.                                                                                                                                                          |
| 251 | <code>scored_results.sort(</code>                          | Sorts <code>scored_results</code> in place.                                                                                                                                                        |
| 252 | <code>key=lambda item: item["score"],</code>               | Sort key: a lambda that reads the <code>score</code> value from each result dictionary.                                                                                                            |
| 253 | <code>reverse=True</code>                                  | <code>reverse=True</code> puts the highest score first (descending order).                                                                                                                         |
| 254 | <code>)</code>                                             | Closes the <code>sort(</code> call.                                                                                                                                                                |
| 255 |                                                            | Blank line before the return.                                                                                                                                                                      |
| 256 | <code>return scored_results[:top_k]</code>                 | Returns only the first <code>top_k</code> results ( <code>[:top_k]</code> ). In the RAG flow, these file paths, snippets and scores are the retrieved context passed on to the LLM/analysis stage. |

**Flow summary (lines 18 to 256):** `_tokenize` (text to words) → `build_simple_rag_index` (project files to in-memory documents) → `_calculate_match_score` (token-overlap score) and `_make_snippet` (preview text) → `query_rag_engine` (phrase boost, ranking, top-k files for the LLM). All 256 lines of the file are covered above, including blank lines.